Pierre Beckmann

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Education

2025 – Now **PhD student in computer science**, EPFL and Idiap Research Institute
Neuro-symbolic AI group of Prof. André Freitas, M-RATIONAL project; reasoning and mechanistic interpretability in language models.

2019 – 2023 **M.A. in Philosophy**, University of Lausanne
Preceded by a propaedeutic year and 40 catch-up credits.
Philosophy of science, Kantian philosophy and phenomenology; exchange semester at Humboldt University Berlin in 2022. Thesis: *Le transcendantal, le physiologique et le computationnel: le predictive processing à partir de Kant et de Helmholtz*.

2016 – 2019 **M.Sc. in Computational Science and Engineering**, ETH Zurich
Machine learning, NLP, computer vision and HPC. Thesis: *Story understanding and story generation with deep learning*.

2013 – 2016 **B.Sc. in Life Sciences and Technologies**, EPFL

Research experience

2026 – Now **MATS Scholar**, Machine Learning Alignment & Theory Scholars, Berkeley
Supervised by Patrick Butlin. Extension phase since May 2026.

2023 – 2024 **Doctoral researcher in philosophy of AI**, University of Bern
Research in the philosophy of deep learning, supervised by Prof. Claus Beisbart.

2019 – 2021 **Research engineer (50%)**, EPFL LSIR
Implemented and deployed multimodal-LLM-based video retrieval systems.

2019 – 2023 **Research engineering intern, then scientific advisor (10%)**, Logitech, Lausanne
Deep learning for audio: first product-feasible research, then supervision of interns and master students, resulting in joint publications.

2018 – 2019 **Master thesis and semester project**, Disney Research Zurich
Story understanding and generation with deep learning.

Teaching experience

2024 **Assistant lecturer, Philosophy of AI**, University of Bern
Co-taught two CAS courses in Advanced Machine Learning, delivering one third of the lectures: *Philosophy and Ethics of Extended Cognition and Artificial Intelligence*; *NLP: Philosophical and Ethical Aspects*.

2023 **Tutor, Philosophy of Mind**, University of Lausanne
Introductory course in contemporary philosophy of mind for first-year students.

2021 – 2022 **Tutor, General and Systematic Philosophy**, University of Lausanne
Introductory course in phenomenology for first-year students; designed and ran three methodology workshops.

Publications in philosophy

Published **Where is the Mind? Persona Vectors and LLM Individuation**
Pierre Beckmann, Patrick Butlin.
Ergo, forthcoming.

Mechanistic Indicators of Understanding in Large Language Models
Pierre Beckmann, Matthieu Queloz.
Philosophical Studies, 2026.

New Horizons in Machine Understanding: Explanatory and Objectual Understanding in Deep Learning Video Generation Models
Pierre Beckmann.
Synthese, 2025.

An alternative to cognitivism: computational phenomenology and deep learning
Pierre Beckmann, Guillaume Köstner, Ines Hipolito.
Minds and Machines, 2023.

Under review **A Polyphonic Conception of AI Understanding**
Matthieu Queloz, **Pierre Beckmann**.
arXiv, 2026.

Why We Care About Understanding: Competence through Predictive Compression
Matthieu Queloz, **Pierre Beckmann**.
PhilPapers preprint, 2025.

Publications in LLM research

Published **SciR: A Controllable Benchmark for Scientific Reasoning in LLMs**
Pierre Beckmann, Marco Valentino, André Freitas.
NeurIPS, 2026.

Probing Persona-Dependent Preferences in Language Models
Oscar Gilg, **Pierre Beckmann**, Daniel Paleka, Patrick Butlin.
NeurIPS, 2026.

Adaptive LLM-Symbolic Reasoning via Dynamic Logical Solver Composition
Lei Xu, **Pierre Beckmann**, Marco Valentino, André Freitas.
EACL, 2026.

Under review **World Modeling in Transformers**
Pierre Beckmann, Matthieu Queloz, André Freitas.
arXiv, 2026.

Publications in deep learning for speech

No longer active in this field.

Published **BYOL-S: Learning self-supervised speech representations by bootstrapping**
Gasser Elbanna, Neil Scheidwasser-Clow, Mikolaj Kegler, **Pierre Beckmann**, Karl El Hajal, Milos Cernak.
PMLR, 2023; 3rd place in the HEAR benchmark.

Hybrid Handcrafted and Learnable Audio Representation for Analysis of Speech Under Cognitive and Physical Load
Gasser Elbanna, Alice Biryukov, Neil Scheidwasser-Clow, Lara Orlandic, Pablo Mainar, Mikolaj Kegler, **Pierre Beckmann**, Milos Cernak.
Interspeech, 2022.

SERAB: A multi-lingual benchmark for speech emotion recognition
Neil Scheidwasser-Clow, Mikolaj Kegler, **Pierre Beckmann**, Milos Cernak.
ICASSP, 2021.

Word-Level Embeddings for Cross-Task Transfer Learning in Speech Processing
Pierre Beckmann, Mikolaj Kegler, Milos Cernak.
EUSIPCO, 2021.

Deep speech inpainting of time-frequency masks
Pierre Beckmann, Mikolaj Kegler, Milos Cernak.
Interspeech, 2020.

Course book

Neural Networks for Cognitive Sciences
2025. Three chapters, with Prof. Jeff Yoshimi, University of California Merced: history of neural networks; convolutional neural networks; transformers and LLMs.

Talks

Nov. 2026. Talk at The Age of the Bookworm (upcoming)
University of Geneva

Oct. 2026. A map of Manhattan inside of TaxiGPT
Idiap Research Institute, Martigny

July 2026. Mechanistic indicators of understanding in LLMs
Cultural Analytics and Digital Approaches in Philosophy, online seminar

June 2026. Building a reflective equilibrium machine?
Workshop, University of Bern

June 2026. Faut-il se soucier du bien-être des IAs?
Kiosque de Philosophie, Groupe Vaudois de Philosophie, Lausanne

Jan. 2026. Do LLMs understand? A mechanistic perspective
Invited talk, CIS Brownbag Speaker Series, University of California, Merced

Nov. 2025. Why we care about understanding: competence via predictive compression
Philosophy of science colloquium, University of Bern

Oct. 2025. Information recall in deep learning: beyond the feature combination paradigm
PhAI 2025, Vrije Universiteit Amsterdam

Feb. 2025. Computational phenomenology and deep learning
Calculating Experience? seminar, St Catherine's College, Oxford

Nov. 2024. Les neurosciences computationnelles à l'épreuve de la méthode transcendantale
Kant tercentenary series, University of Lausanne

Oct. 2024. Could deep learning video generation models understand the physical world?
Idiap Research Institute

Nov. 2022. Computational phenomenology and deep learning
Series on AI and consciousness, University of Lausanne

Reviewing

Journals
Ergo, Philosophical Studies, Synthese, Res Philosophica, Journal of Consciousness Studies, Knowledge in Society.

Workshops and other
Mechanistic Interpretability Workshop (ICML 2026); MathNLP Workshop (EMNLP 2025, 2026); Digital Minds Research Sprint (2026); Ubiquity Press, *Understanding Conversational AI* (2025).

Grants

2026 AI Safety Support: stipend of USD 36'000 for the MATS extension
2026 AI Safety Support: stipend of USD 14'400 for the MATS program
2023 University of Bern: seed funding of CHF 10'000 for a research period

Service

2026 – Now Member and co-founder of PhilInterp, the Philosophy with Mechanistic Interpretability Network
2024 – Now Committee member, Groupe Vaudois de Philosophie